

ORDER NUMBER	0179784
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MATERIAL TEST REPORT

S O L D	DODSON GLOBAL INC P O BOX 330759 HOUSTON TX 77233	S H I P	DODSON GLOBAL 2750 HOLMES ROAD HOUSTON TX 77051
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ITEMS

Ln#	Qty	Description	Matl Spec	Ht Cd	Mill Heat Number	CE
009	4	8" 600 RF WN XH MSS A694 F65	A694H-16/MSS-SP-44-19	1E116H	NU4000016461	0.46
018	5	8" 900 RF WN STD MSS A694 F52	A694N-16/MSS-SP-44-19	1E043N	NU4000015328	0.47

CHEMICAL PROPERTIES

* Ladle Chemical Properties Indicated with an Asterisk (*)

Ht Cd	C	Si	Mn	P	S	Cr	Al	Cu	Ni	Mo	V	Nb	Sn	N	B	Pb	Ti
1E116H	0.2000	0.2580	1.2200	0.0138	0.0073	0.1740	0.0301	0.1360	0.0633	0.0391	0.0518	0.0005	0.0000	0.0000	0.0003	0.0000	0.0020
*	0.1800	0.2580	1.2700	0.0110	0.0050	0.1700	0.0240	0.1300	0.0600	0.0400	0.0580	0.0010	0.0100	0.0000	0.0002	0.0000	0.0010
1E043N	0.1900	0.2680	1.2400	0.0054	0.0025	0.1330	0.0282	0.1860	0.0600	0.0189	0.0639	0.0005	0.0000	0.0000	0.0005	0.0000	0.0021
*	0.2000	0.2770	1.2500	0.0080	0.0010	0.1300	0.0230	0.1900	0.0600	0.0400	0.0560	0.0030	0.0040	0.0000	0.0006	0.0000	0.0020

PHYSICAL PROPERTIES

* Tensile and Yield Strengths are shown in psi

Ht Cd	Tensl*	Yield*	%Elong	%R.A.	HBW	Charpy	Ft-Lbs	Lat Expsn	%-Shear	Test Temp
1E116H	98000	75000	29.30	68.30	217/221	TRANSV.	38/40/37	.020"/.017"/.017"	40/30/30	-50 F
1E043N	81500	52200	34.50	69.48	170/170	TRANSV.	75/39/42	.049"/.021"/.026"	45/20/20	-50 F

NOTES

Ht Cd	Country of Melt	Note
1E116H	United States	1650F 1HR/INCH, WATER QUENCH, TEMPER 1000F 1HR/INCH, AIR COOL
1E043N	United States	NORMALIZED, 1650F FOR 1HR/INCH, AIR COOL

All flanges bearing the "BOLTEX" name are machined in the United States at our Houston Texas facility. Flanges marked with "MADE IN USA" or "USA" are made from steel melted in the United States. Country of melt is shown in the notes section above. Billet used in the manufacturing process has been fully killed. Boltex flanges are manufactured, sampled, tested and inspected per MSS SP-44 2019 PSL1.

It is guaranteed that all Boltex forged steel flanges marked B16 or MSS, adhere to and comply with all dimensions and tolerances in accordance with specification ASME B16.36-2020, ASME B16.5-2020/ASME B16.47-2020 and MSS SP-44-2019 PSL1 as applicable for flanges. Boltex flanges have a surface finish of 125/250 roughness average (Ra) on all full face and raised face finishes.

We certify our flanges marked B16 or MSS, are capable of passing a hydrostatic test compatible with their rating and all test results and process information contained herein are correct and true as contained in company records. The flanges meet the requirements of the latest edition of NACE MR0175/ISO15156 and MR0103/ISO17945.

Our quality system has been registered by DNV-GL to ISO 9001, Certificate No.: 142240-2013-AQ-USA-ANAB. This MTR complies with EN 10204-3.1. Our flanges satisfy the material requirements for PED 2014/68/EU ANNEX 1, Section 4.3, Type 3.1. Notwithstanding the absence of a signature, the organization submitting either a printed certificate or an EDI transmitted certificate is responsible for report content or the report as per ASTM A961-21 Section 19.5.

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Quality Assurance Manager